

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ac64db9a07f16b324.jsonl`  
- SHA-256: `0110b65611bb8b62c618de20fbfd8f5c8660c32e9c4a63cc8001a5665d1386d9`  
- Source modified: `2026-06-16T03:24:56+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:24:13.013Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: ****What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?**** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best ***recent*** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"Detecting action boundaries with a dedicated branch yields large gains specifically on over-segmentation-sensitive metrics: up to 13.7% improvement in segmental edit distance and up to 16.1% in segmental F1 score over prior state-of-the-art."

****Source:**** <https://arxiv.org/abs/2007.06866> (primary)

****Supporting quote:**** "improvements of "up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score" over prior state-of-the-art methods"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:24:17.436Z)

I'll verify this claim about action boundary detection yielding gains on over-segmentation metrics. The source is arXiv 2007.06866. Let me fetch the primary source.

3. user (2026-06-16T03:24:24.858Z)

Paper Summary

Title: "Alleviating Over-segmentation Errors by Detecting Action Boundaries"

Yes, the paper reports those specific improvements. According to the abstract, it achieves "an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score."

Proposed Method: The paper introduces the Action Segment Refinement Framework (ASRF), which uses two specialized components. One branch handles frame-wise action classification, while a second branch performs boundary regression. The framework leverages "predicted action boundaries" to refine the frame-level classifications.

The authors also propose "a loss function for smoothing the transition of action probabilities" as part of their contribution to temporal action segmentation.

4. assistant (2026-06-16T03:24:26.938Z)

The primary source confirms the claim directly. Let me verify this is the well-known ASRF paper and check the publication venue and any contradicting evidence.

5. user (2026-06-16T03:24:32.854Z)

Web search results for query: "ASRF "Alleviating Over-segmentation Errors by Detecting Action Boundaries" WACV 2021 segmental edit F1"

Links: [{"title": "asrf/evaluate.py at main · yiskw713/asrf", "url": "https://github.com/yiskw713/asrf/blob/main/evaluate.py"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "GitHub - yiskw713/asrf: Code for 'Alleviating Over-segmentation Errors by Detecting Action Boundaries' accepted in WACV2021", "url": "https://github.com/yiskw713/asrf"}, {"title": "WACV 2021 Open Access Repository", "url": "https://openaccess.thecvf.com/content/WACV2021/html/Ishikawa_Alleviating_Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.html"}, {"title": "Alleviating Over-Segmentation Errors by Detecting Action Boundaries", "url": "https://papertalk.org/papertalks/27315"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/9423184/?jsessionid=AF16C06E2134F234A6ACC2E82774FD19"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://arxiv.org/pdf/2007.06866"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | Request PDF", "url": "https://www.researchgate.net/publication/352386005_Alleviating_Over-segmentation_Errors_by_Detecting_Action_Boundaries"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://openaccess.thecvf.com/content/WACV2021/papers/Ishikawa_Alleviating_Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.pdf"}, {"title": "Alleviating over-segmentation errors by detecting action boundaries ? Keio University", "url": "https://keio.pure.elsevier.com/en/publications/alleviating-over-segmentation-errors-by-detecting-action-boundari-2"}]

I found the paper you're looking for. Here are the key findings:

The paper "Alleviating Over-Segmentation Errors by Detecting Action Boundaries" was published in WACV 2021 by Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, and Hirokatsu Kataoka.

Regarding the segmental edit F1 metric you asked about: The model outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 16.1% in terms of segmental F1 score.

The framework, called Action Segment Refinement Framework (ASRF), consists of a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB). The model also offers an improvement of up to 13.7% in terms of segmental edit distance.

The official code for the paper is publicly available on GitHub at <https://github.com/yiskw713/asrf>.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:24:46.986Z)

The claim is fully corroborated. This is the ASRF paper, peer-reviewed and published at WACV 2021, a reputable venue. The abstract explicitly states the exact numbers cited in the claim. Multiple independent sources (arXiv, CVF Open Access, IEEE Xplore, the official GitHub, the authors' project page, Keio University) confirm both the method (Boundary Regression Branch for action boundary detection) and the specific numbers.

Let me assess against the checklist:

1. ****Supported by quote?**** Yes ? the abstract verbatim says "up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score." The claim accurately reports this. The framing ("specifically on over-segmentation-sensitive metrics") is also accurate: segmental edit distance and segmental F1 are precisely the metrics designed to penalize over-segmentation, which is the paper's stated focus.
2. ****Contradicting evidence?**** None found. The paper is well-cited and the method (ASRF) is a recognized baseline in temporal action segmentation.
3. ****Source quality?**** Strong ? WACV 2021 (peer-reviewed CVF/IEEE venue), primary source, with public code. Matches the claim strength.
4. ****Outdated?**** Published 2020/2021. Within the broader TAL literature this is a foundational, still-cited method. Not stale for the purpose cited (it's a canonical over-segmentation method).
5. ****Marketing/cherry-pick?**** No ? these are standard benchmark results on three public datasets (50Salads, GTEA, Breakfast).

One minor nuance worth flagging: "up to" means these are best-case improvements across datasets/metrics, not uniform gains ? but the claim faithfully uses "up to," so it is not an overreach.

7. user (2026-06-16T03:24:53.365Z)

Structured output provided successfully

8. assistant (2026-06-16T03:24:56.390Z)

Verification complete. The claim is ****not refuted**** ? it is fully supported by the primary peer-reviewed source (ASRF, WACV 2021, arXiv:2007.06866). The exact numbers (13.7% segmental edit distance, 16.1% segmental F1) appear verbatim in the abstract, the "up to" qualifier is faithfully preserved, and the over-segmentation framing matches the paper's stated purpose. Confidence: high.